

Cyber Attack Detection in P2P Energy Trading Using Machine Learning

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Abstract—A system in which consumers trade electricity directly is called P2P energy trading. P2P energy trading is attracting attention as a new power system, but measures against cyber attacks are important. In this paper, we discuss cyber attack detection in P2P energy trading. We apply several machine learning methods to real data and compare their accuracy.

Index Terms—P2P energy trading, cyber attack detection, machine learning

I. INTRODUCTION

Peer-to-peer (P2P) energy trading has emerged as a new energy management model in which network participants, known as prosumers, share a portion of their energy resources with one another without the direct involvement of a central controller [1]–[4]. One key benefit of a P2P trading platform is that prosumers in need of energy can purchase surplus energy from others in the same network at a lower price compared to the main grid's selling price.

Preserving data integrity is fundamental to ensuring the security of the energy trading market. Integrity breaches can occur due to the corruption of data exchanges resulting from attacks such as false data injection attacks (FDIAs), replay attacks, and man-in-the-middle attacks.

In this paper, we discuss cyber attack detection in P2P energy trading systems. We design a P2P energy trading model involving prosumers, suppliers, and an attacker who falsifies prosumer demand data to gain free energy. Several machine learning methods, including SVM, NB, DT, KNN, RF, MLPNN, and XGBoost, are applied to detect such cyber attacks. Furthermore, we propose an optimization strategy that uses Particle Swarm Optimization (PSO) to fine-tune the hyperparameters of the XGBoost classifier. The effectiveness of the optimized model is validated and compared with other methods using real-world data.

II. P2P ENERGY TRADING

A community-based P2P market model is designed with four types of agents: prosumers (including pure consumers and prosumers with solar generation), suppliers (with energy generation from solar, wind, or conventional power plants), a coordinator, and an attacker. Both suppliers and prosumers aim to maximize their profits in a non-cooperative game setting.

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Prosumers determine the amount of energy to buy or sell each hour based on their solar generation and consumption, and send their demand data to the coordinator. Suppliers produce energy and sell it to prosumers when local generation is insufficient. Meanwhile, the attacker intercepts and falsifies some prosumer demand data.

The attacker hacks into a prosumer's home energy management (HEM) system early in the market cycle and falsifies shiftable and base loads to distort demand. The manipulated data is submitted with other honest reports. The coordinator computes net load, receives supplier bids, and iteratively updates prices. The attacker continuously injects false data to create exploitable surplus energy until the market stabilizes.

III. PROPOSED CLASSIFICATION METHOD

To detect whether the data has been attacked, we propose a particle swarm optimization-based parameter selection method for XGBoost. The optimized XGBoost classifier is used for classification.

Particle Swarm Optimization (PSO) is particularly useful for optimizing continuous functions. In PSO, each candidate solution is represented as a particle in the search space. These particles move around the space to find the optimal solution based on their own experience and the experience of their neighbors. The goal is to use Particle Swarm Optimization to find the optimal hyperparameters for an XGBoost model to achieve better predictive performance. PSO is used to automatically search for the best combination of parameters such as `n_estimators`, `max_depth`, `learning_rate`, `subsample`, and `colsample_bytree` that maximize the models' accuracy on a validation set.

PSO Mechanism: Particles: Each particle represents a potential solution in the search space. The position of a particle corresponds to a possible set of hyperparameters for XGBoost.

• **Velocity Update:** Particles adjust their positions based on a velocity that is updated in each iteration using the following formula: $v_i(t+1) = w \cdot v_i(t) + c_1 \cdot r_1 \cdot (p_i - x_i(t)) + c_2 \cdot r_2 \cdot (g - x_i(t))$, where: $v_i(t)$: Velocity of particle i at time t . w : Inertia weight, controlling the influence of the previous velocity. c_1 : Cognitive coefficient, guiding the particle towards its personal best position p_i . c_2 : Social coefficient, guiding the particle towards the global best position g . r_1, r_2 : Random numbers

between 0 and 1, introducing stochastic behavior. $x_i(t)$: Current position of particle i at time t .

• **Position Update:** The position of each particle is updated as $x_i(t+1) = x_i(t) + v_i(t+1)$, where $x_i(t+1)$ is the new position of the particle i after the velocity update.

Nonlinear Decreasing Inertia Weight: The inertia weight w controls the balance between exploration (global search) and exploitation (local search). A nonlinear decreasing inertia weight allows the algorithm to focus on global exploration in the initial stages and switch to local exploitation in later stages. It can be updated using:

$$w(i) = w_{\text{end}} + \left(\frac{\text{imax} - i}{\text{imax}} \right)^n \cdot (w_{\text{ini}} - w_{\text{end}}), \quad (1)$$

where w_{ini} : Initial inertia weight (e.g., 0.9) and w_{end} : Final inertia weight (e.g., 0.4).

Asynchronous Linearly Varying Learning Factors: The learning factors c_1 and c_2 change linearly over time to balance the influence of personal best and global best positions [5]:

$$c_1(i) = c_{1\text{ini}} - (c_{1\text{ini}} - c_{1\text{fin}}) \cdot \frac{i}{\text{imax}} \quad (2)$$

$$c_2(i) = c_{2\text{ini}} + (c_{2\text{fin}} - c_{2\text{ini}}) \cdot \frac{i}{\text{imax}} \quad (3)$$

where $c_{1\text{ini}}$ and $c_{2\text{ini}}$: Initial values of the learning factors, and $c_{1\text{fin}}$ and $c_{2\text{fin}}$: Final values of the learning factors.

Objective Function: The objective function measures the quality of a set of XGBoost parameters based on cross-validation accuracy:

$$\text{Objective}(\text{param}) = -\frac{1}{k} \sum_{j=1}^k \text{Accuracy}(\text{param}, \text{fold}_j), \quad (4)$$

where param : A vector of XGBoost parameters (e.g., `n_estimators`, `max_depth`, etc.), k : Number of cross-validation folds, and $\text{Accuracy}(\text{param}, \text{fold}_j)$: Accuracy obtained on the j -th fold using the parameters param .

IV. NUMERICAL EXAMPLE

We use real data from Austin, Texas (Aug. 1, 2018) [6] including six features (ID, generation, shiftable/base load, consumption, demand). Attack data is created by falsifying load values. The test set uses July 1 data. Group 1 includes 50 households (20 prosumers with PV, 1 attacker), and Group 2 has 23 households (with 1 network-level attacker). Three suppliers participate in the market.

The P2P system model contains two groups of households: Group 1: Contains 50 households, with 20 prosumers equipped with rooftop PV panels (one of them is the attacker) and 30 consumers with zero energy generation, Group 2: Includes 23 households (13 prosumers and 10 consumers) and one attacker who intercepts prosumer/coordinator communication at the network level. Three companies act as suppliers in this P2P market.

We compare the proposed method with other machine learning algorithms including Support Vector Machine (SVM), Naive Bayes (NB), Decision Tree (DT), K-Nearest Neighbors

Table. I
COMBINED RESULTS

Model	AR	Precision	Recall	f1-score
GROUP 1				
Proposed method	99.1	97.2	98.6	97.9
SVM	90.6	85.6	67.9	75.7
NB	86.0	75.3	52.1	61.6
DT	95.7	97.8	96.7	97.2
KNN	95.2	88.7	89.3	89.0
RF	98.2	97.8	93.6	95.6
MLPNN	95.2	92.9	84.3	88.4
XGboost	98.4	96.4	96.4	96.4
GROUP 2				
Proposed method	98.0	95.4	97.6	96.5
SVM	95.0	93.8	88.2	90.9
NB	84.2	95.2	47.1	63.0
DT	96.6	92.1	96.5	94.2
KNN	92.6	88.9	84.7	86.7
RF	96.6	94.1	94.1	94.1
MLPNN	95.6	91.9	92.9	92.4
XGboost	97.3	95.3	95.3	95.3

(KNN), Random Forest (RF), Multilayer Perceptron Neural Network (MLPNN), and XGBoost. In the context of classification, the following terms are used to describe the performance of a model: True Positive (TP), True Negative (TN), False Positive (FP), False Negative (FN). We consider the following four performance metrics: (i) Accuracy = $(TP + TN) / (TP + TN + FP + FN)$, (ii) Precision = $TP / (TP + FP)$, (iii) Recall = $TP / (TP + FN)$, and (iv) F1-Score = $2(\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$.

As shown in Table 1, the proposed method consistently achieves the best or near-best performance across all datasets, with high accuracy and a balanced trade-off between precision and recall.

V. CONCLUSION

This paper proposed a method for detecting cyber attacks in P2P energy trading systems using an optimized XGBoost model, where parameters are fine-tuned through particle swarm optimization. By a numerical example, we demonstrated the superior performance of the proposed method.

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